

Uttam Kumar Mahto

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LinkedIn — GitHub — Portfolio

PROFESSIONAL SUMMARY

Backend-Focused Full Stack Engineer with **3+ years of experience** designing and developing **scalable, high-performance web applications** across Healthcare, EdTech, and RetailTech domains. Strong expertise in **Node.js, Express.js, React.js, Ruby on Rails, REST API development, PostgreSQL, MySQL, Redis, Elasticsearch, Docker, AWS, CI/CD**, and **system optimization**. Experienced in building **multi-tenant systems, authentication and authorization modules, distributed backend services, search and data retrieval systems, and cloud-native applications**. Hands-on knowledge of **AI/LLM application development**, including **Retrieval-Augmented Generation (RAG), embeddings, vector search with PostgreSQL/pgvector, prompt engineering, AI chatbots**, and **LLM API integration using OpenAI and Gemini**.

TECHNICAL SKILLS

Programming Languages: JavaScript (ES6+), TypeScript, Ruby, SQL

Frontend Technologies: React.js, Next.js, Vue.js, Redux, React Query, Tailwind CSS, HTML5, CSS3

Backend Technologies: Node.js, Express.js, Ruby on Rails, REST APIs, Authentication, Authorization, RBAC

Databases: PostgreSQL, MySQL, MongoDB, Redis

Search & Data Retrieval: Elasticsearch, Full-Text Search, Fuzzy Search, Semantic Search, Vector Search, Relevance-Based Retrieval

Cloud & DevOps: AWS EC2, AWS S3, Docker, Jenkins, Drone CI, Nginx, CI/CD Pipelines

Testing Tools: Cypress, Jest, Mocha, Chai, Postman

AI & LLM Technologies: LLMs, RAG, Prompt Engineering, Embeddings, Vector Databases, Semantic Search, AI Chatbots, OpenAI API, Gemini API, Claude

Monitoring & Observability: Grafana, New Relic

Core Concepts: Scalable Systems, Distributed Systems, Multi-Tenant Architecture, API Optimization, Performance Tuning, System Design, Microservices, Search Systems, RAG, LLM Applications, Containerization, Cloud Deployment

Tools: Git, GitHub, GitLab, Jira, Claude Code, GitHub Copilot

PROFESSIONAL EXPERIENCE

Full Stack Developer

Feb 2026 – Present

Sanpbizz CloudTech Pvt. Ltd. – Bangalore, Karnataka

Retailer and Warehouse Management Platform

- Designed and developed **scalable backend services** for retailer and warehouse management systems using **Node.js and Express.js**.
- Built **RESTful APIs** for inventory tracking, order lifecycle management, stock reconciliation, and warehouse operations.
- Implemented secure **Role-Based Access Control (RBAC)** for retailers, warehouse managers, and administrators.
- Designed **multi-tenant architecture** ensuring retailer-level data isolation and secure access management.
- Implemented **Elasticsearch-based product and inventory search** with **fuzzy matching and fuzziness: AUTO** to handle misspellings, partial matches, and user search variations.
- Optimized **search relevance and response performance** by indexing frequently queried product and inventory fields in Elasticsearch.
- Implemented **Redis-based caching** for frequently accessed product, inventory, and application data, reducing repeated database queries and improving API response performance.
- Optimized **MySQL queries and API response performance** for high-volume transactional workflows.
- Integrated enterprise workflows for clients including **IRCTC and Axis Bank** retail operations.
- Developed **AI-powered chatbot capabilities using LLM APIs** to enable natural-language interaction with application data and business workflows.
- Implemented **Retrieval-Augmented Generation (RAG)** workflows using **embeddings and PostgreSQL with pgvector** for semantic similarity search and retrieval of relevant domain-specific context.
- Applied **prompt engineering and context management** techniques to improve chatbot response relevance and grounding.
- Containerized backend applications using **Docker** and supported **AWS cloud deployments**.

Environment: Node.js, Express.js, React.js, Vue.js, MySQL, PostgreSQL, Redis, Elasticsearch, Prisma ORM, pgvector, Docker, AWS, LLMs, RAG, OpenAI API, Gemini API

Full Stack Developer

Nov 2024 – Jan 2026

INCANUS Technologies Pvt. Ltd. – Bangalore, Karnataka

Online Exam Hosting and Evaluation Platform

- Designed **backend architecture** for online examination systems supporting **thousands of concurrent users**.

- Developed **scalable REST APIs** for exam creation, coding submissions, automated evaluation, and result generation.
- Integrated **VS Code-based online coding playground** for real-time programming assessments.
- Improved **API performance** and optimized database queries to handle peak traffic efficiently.
- Implemented **authentication, authorization, and RBAC modules** for students, mentors, and administrators.

Environment: Node.js, Express.js, React.js, Vue.js, MySQL, Docker, AWS EC2

CI/CD Automation and Quality Engineering

- Implemented **CI/CD pipelines using Drone CI and Docker** for automated testing and deployments.
- Developed **end-to-end test suites using Cypress** and backend unit tests using **Mocha and Chai**.
- Reduced deployment failures and improved release reliability through **automated deployment workflows**.

Software Engineer

Feb 2022 – Aug 2023

Cerner Healthcare Solutions (Oracle) – Bangalore, Karnataka

HealthIntent Healthcare Platform

- Developed **backend services** for medication management and audit logging systems using **Ruby on Rails**.
- Owned backend modules including **Activity Logs and Medication Workflow services**.
- Built and optimized **React.js components** for clinician dashboards and healthcare workflows.
- Improved **frontend responsiveness and API performance** through query optimization and component tuning.
- Monitored production systems using **Grafana and New Relic** for observability and incident management.
- Configured **Jenkins pipelines and Docker-based deployments** enabling zero-downtime releases.
- Collaborated with QA and DevOps teams to improve **production stability and deployment reliability**.

Environment: Ruby on Rails, Node.js, React.js, PostgreSQL, Redis, Docker, Jenkins

PROJECTS

Scalable Coding Assessment Platform

Node.js, React.js, Docker

- Developed a **full-stack coding assessment platform** supporting MCQs and real-time coding challenges.
- Implemented **authentication, authorization, RBAC, automated code evaluation**, and reusable React components.
- Designed **scalable REST APIs** and Dockerized backend services for deployment and scalability.

Authentication and Session Management Service

Node.js, Express.js, MySQL

- Built secure **JWT authentication and refresh token workflows** for scalable web applications.
- Implemented **protected routes, middleware-based authorization, token expiry handling, and API security**.
- Added **unit and integration testing** using Mocha and Chai.

Workflow Automation and Notification System

Node.js, Twilio API, SMTP

- Developed **automated workflow management system** with email and WhatsApp notification integrations.
- Integrated **Twilio APIs and SMTP services** for real-time alerts and approval workflows.
- Designed **backend APIs and asynchronous communication pipelines** for scalable request processing.

RAG-Powered Invoice and Product Analytics Chatbot

Node.js, pgvector, Elasticsearch

- Developed an **AI-powered RAG chatbot** using **Node.js** to interact with invoice, product, and business datasets through natural-language queries.
- Implemented **hybrid search** combining **Elasticsearch keyword/fuzzy search** with **pgvector semantic vector search** to improve retrieval across structured and unstructured business data.
- Built an **embedding-based retrieval pipeline** using PostgreSQL and **pgvector** to identify semantically relevant products, invoices, and business records.
- Implemented **Elasticsearch fuzzy matching and relevance-based retrieval** to handle misspellings, partial product names, SKU variations, and keyword-based queries.
- Designed a **hybrid retrieval and ranking workflow** that combines lexical and semantic search results before passing relevant context to the LLM.
- Enabled users to ask natural-language questions about **invoice history, product information, sales data, purchasing patterns, and business analytics**.
- Implemented **RAG context construction** to retrieve relevant records and provide grounded context to the LLM, reducing reliance on model-generated assumptions.
- Designed backend APIs for **dataset ingestion, embedding generation, indexing, search, retrieval, and chatbot interactions**.

Environment: Node.js, Express.js, PostgreSQL, pgvector, Elasticsearch, Redis, LLM APIs, RAG, Embeddings, Hybrid Search, REST APIs

EDUCATION

Master of Computer Applications (MCA)

2022

Reva University – Bangalore, Karnataka

Bachelor of Science in Information Technology

2019

Marwari College – Ranchi, Jharkhand